THE MODE OF CLASSICAL AND FREE BROWNIAN MOTION WITH AN INITIAL DISTRIBUTION AT LARGE TIME

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Abstract. Let μ be a probability measure on the real line with mean m and finite variance v. We show that the density of $\mu * N(0,t)$ has a unique global maximizer for $t \geq 600v$ and that the density of the free convolution $\mu \boxplus S(0,t)$ with the semicircle law of variance t has a unique global maximizer for $t > 7(7+4\sqrt{2})v$. Both maximizers converge to m as $t \to \infty$, with explicit bounds of order $t^{-1/2}$, and of order t^{-1} when the third absolute moment is finite. If the seventh moment is finite, the maximizers are

$$m - \frac{\mu_3}{2t} + \frac{\mu_5 - 6v\mu_3}{8t^2} + O(t^{-3}) \qquad \text{and} \qquad m - \frac{3\mu_3}{t} + \frac{5\mu_5 - 22v\mu_3}{t^2} + O(t^{-3})$$

respectively, where μ_k is the kth central moment of μ. This gives an answer to a problem of Hasebe and Ueda on the position of the mode of classical and free Brownian motion with an initial distribution.

1. Introduction

Let μ be a probability measure on $\mathbb{R}$. The law at time t of a Brownian motion with initial distribution μ is the convolution $\mu * N(0,t)$, and the law at time t of a free Brownian motion with initial distribution μ is the free convolution $\mu \boxplus S(0,t)$, where $S(0,t)$ is the semicircle law with mean 0 and variance t, whose density is $(2\pi t)^{-1}\sqrt{4t - x^2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$. Free convolution of arbitrary probability measures on $\mathbb{R}$ is defined by Bercovici and Voiculescu [1], and the measures $\mu \boxplus S(0,t)$ are described by Biane [2]: each of them has a continuous density.

A probability measure on $\mathbb{R}$ is unimodal if, for some $a \in \mathbb{R}$, it is the sum of an atom at a and a measure with a density that is non-decreasing on $(-\infty, a)$ and non-increasing on (a, ∞); such a point a is a mode. Hasebe and Ueda [8] study the unimodality of $\mu * N(0,t)$ and $\mu \boxplus S(0,t)$ for large t. They prove that if

$$\alpha := \int_{\mathbb{R}} e^{\varepsilon x^2}\, d\mu(x) < \infty \qquad \text{for some } \varepsilon > 0, \tag{1}$$

then $\mu * N(0,t)$ is unimodal for all $t \geq 36\log(2\alpha)/\varepsilon$ [8, Theorem 4.3], and that if μ has compact support of diameter D, then $\mu \boxplus S(0,t)$ is unimodal for all $t \geq 4D^2$ [8, Theorem 3.2]. Neither statement extends to all μ: there are measures μ for which $\mu * N(0,t)$ is not unimodal for any $t > 0$ [8, Proposition 4.2], and for every Borel function $f\colon \mathbb{R} \to [0,\infty)$ there is a measure μ with $\int f\, d\mu < \infty$ for which $\mu \boxplus S(0,t)$ is not unimodal for any $t > 0$ [8, Proposition 3.4]. In particular, the condition (1) does not imply that $\mu \boxplus S(0,t)$ is eventually unimodal. For unimodal μ the law $\mu * N(0,t)$ is unimodal for every t, because the normal law is strongly unimodal in the sense of Ibragimov [10]; the condition (1) does not require μ to be unimodal. Unimodality in free probability is studied for freely selfdecomposable laws by Hasebe and Thorbjørnsen [7], for free Lévy processes by Hasebe and Sakuma [6], and, as log-unimodality, for free positive multiplicative Brownian motion by Hasebe, Ueda and Wang [9]. The paper [8] closes its section on the classical case with the following problem.

Date: November 2026.

Problem ([8, Problem 4.6]). *Estimate the position of the mode of classical Brownian motion with initial distributions satisfying the assumption* (1). *Our proof shows that for* $t \geq 36\log(2\alpha)/\varepsilon$, *the mode is located in the interval* $[-\sqrt{t}/2, \sqrt{t}/2]$. *How about free Brownian motion?*

We consider the global maximizers of the two densities. For a unimodal law with a continuous density that has a unique global maximizer, this maximizer is the unique mode; for a law that is not unimodal it is still defined, and it is the natural object in the free case because of [8, Proposition 3.4]. A finite variance is enough for the global maximizer to be unique at large times and to converge to the mean of μ.

Throughout, X is a random variable with law μ and finite variance, and

$$m = \mathbb{E}X, \qquad Y = X - m, \qquad v = \mathrm{Var}(X), \qquad \mu_k = \mathbb{E}Y^k, \qquad \beta_k = \mathbb{E}|Y|^k.$$

If $v = 0$, then $\mu * N(0,t) = N(m,t)$ and $\mu \boxplus S(0,t)$ is the semicircle law of variance t centred at m, and both densities have the unique maximizer m for every $t > 0$. We therefore assume $v > 0$.

Theorem 1 (classical Brownian motion). *Let* $0 < v < \infty$ *and* $t \geq 600v$. *The density of* $\mu * N(0,t)$ *has a unique global maximizer* $M_{\mathrm{c}}(t)$. *It satisfies*

$$|M_{\mathrm{c}}(t) - m| \leq \frac{\sqrt{2}\,v}{\sqrt{t}},$$

and, if $\beta_3 < \infty$,

$$|M_{\mathrm{c}}(t) - m| \leq \frac{\beta_3}{t}.$$

Theorem 2 (free Brownian motion). *Let* $0 < v < \infty$, *let* $a = 7 + 4\sqrt{2}$ *and let* $t > 7av$. *The density of* $\mu \boxplus S(0,t)$ *has a unique global maximizer* $M_{\mathrm{f}}(t)$. *It satisfies*

$$|M_{\mathrm{f}}(t) - m| \leq \frac{4av}{\sqrt{t}},$$

and, if $\beta_3 < \infty$,

$$|M_{\mathrm{f}}(t) - m| \leq \frac{26\big((2+\sqrt{2})^3 v^{3/2} + \beta_3\big)}{t}.$$

Since $v^{3/2} \leq \beta_3$ and $(2+\sqrt{2})^3 = 20 + 14\sqrt{2}$, the last bound is at most $1061\,\beta_3/t$. The numerical constants in the two theorems, and the thresholds $600v$ and $7av \approx 88.6\,v$, are not optimal.

Theorem 3 (expansions). *Let* $0 < v < \infty$.

(a) *If* $\beta_3 < \infty$, *then, as* $t \to \infty$,

$$M_{\mathrm{c}}(t) = m - \frac{\mu_3}{2t} + o(t^{-1}), \qquad M_{\mathrm{f}}(t) = m - \frac{3\mu_3}{t} + o(t^{-1}).$$

(b) *If* $\beta_7 < \infty$, *then, as* $t \to \infty$,

$$M_{\mathrm{c}}(t) = m - \frac{\mu_3}{2t} + \frac{\mu_5 - 6v\mu_3}{8t^2} + O(t^{-3}), \tag{2}$$

$$M_{\mathrm{f}}(t) = m - \frac{3\mu_3}{t} + \frac{5\mu_5 - 22v\mu_3}{t^2} + O(t^{-3}). \tag{3}$$

Thus, when $\beta_3 < \infty$, both maximizers approach the mean at rate t^{-1}; this rate is exact whenever $\mu_3 \neq 0$, and then to leading order the displacement of the free maximizer from the mean is six times that of the classical one. If μ is symmetric about m, then both densities are symmetric about m, and the uniqueness in Theorems 1 and 2 gives $M_{\mathrm{c}}(t) = m$ for $t \geq 600v$ and $M_{\mathrm{f}}(t) = m$ for $t > 7av$.

The expansions take a simple form in terms of cumulants. Let κ_n denote the cumulants of $\mu * N(0,t)$, so that $\kappa_2 = t + v$, $\kappa_3 = \mu_3$ and $\kappa_5 = \mu_5 - 10v\mu_3$. Then (2) reads

$$M_{\mathrm{c}}(t) = m - \frac{\kappa_3}{2\kappa_2} + \frac{\kappa_5}{8\kappa_2^2} + O(t^{-3}). \tag{4}$$

The first term is the classical relation between the mean, the mode and the third cumulant of a nearly normal distribution; see Haldane [4] for distributions with given cumulants and Hall [5] for the mode of a sum of independent random variables. Free cumulants linearize free convolution in the same way as cumulants linearize convolution [15, Lectures 11 and 12]. Let r_n denote the free cumulants of $\mu \boxplus S(0,t)$, so that $r_2 = t + v$, $r_3 = \mu_3$ and $r_5 = \mu_5 - 5v\mu_3$. Then (3) reads

$$M_{\mathrm{f}}(t) = m - \frac{3r_3}{r_2} + \frac{5r_5}{r_2^2} + O(t^{-3}). \tag{5}$$

Asymptotic expansions in the free central limit theorem, for normalized sums of free identically distributed random variables, are due to Chistyakov and Götze [3]. Here the setting is a fixed law perturbed by a semicircular variable of large variance, and (3) is derived directly from Biane's description of the density.

Under the assumptions of [8] the theorems give the following.

Corollary 4. *Let μ satisfy* (1) *and not be a point mass, and put*

$$L = \frac{\log \alpha}{\varepsilon}, \qquad Q = 8\alpha\Big(\frac{3}{2e\varepsilon}\Big)^{3/2}.$$

Then $v \leq L$, $|m| \leq \sqrt{L}$ and $\beta_3 \leq Q$. Consequently:

(i) *for $t \geq \max\{600L,\, 36\log(2\alpha)/\varepsilon\}$, and also for $t \geq 72\log(2\alpha^2)/\varepsilon$, the law $\mu * N(0,t)$ is unimodal, its mode is unique, and the mode lies in $[m - Q/t,\, m + Q/t]$;*

(ii) *for $t > 7(7 + 4\sqrt{2})L$ the density of $\mu \boxplus S(0,t)$ has a unique global maximizer, which lies within $26\big((2+\sqrt{2})^3 L^{3/2} + Q\big)/t$ of m;*

(iii) *the expansions* (2) *and* (3) *hold.*

Corollary 5. *Let μ have compact support of diameter $D > 0$. For $t \geq 23D^2$ the law $\mu \boxplus S(0,t)$ is unimodal, its mode is unique and lies within $136\, D^3/t$ of m, and the mode satisfies* (3).

Theorem 1 is proved in Section 2 and Theorem 2 in Section 3. Section 4 contains the proof of Theorem 3, Section 5 the proofs of the corollaries, and Section 6 an example and numerical checks.

2. The classical case

Replacing X by $X - m$ translates both $\mu * N(0,t)$ and $\mu \boxplus S(0,t)$ by $-m$, so in the proofs of Theorems 1, 2 and 3 we assume $m = 0$; then $Y = X$.

The density of $\mu * N(0,t)$ at h is $(2\pi t)^{-1/2} g(h)$ with

$$g(h) = \mathbb{E}\exp\Big(-\frac{(Y-h)^2}{2t}\Big).$$

The function g is positive and continuously differentiable, and $g'(h) = H(h)/t$, where

$$H(h) = \mathbb{E}\Big[(Y-h)\exp\Big(-\frac{(Y-h)^2}{2t}\Big)\Big].$$

The function H is continuously differentiable, with

$$-H'(h) = \mathbb{E}\Big[\Big(1-\frac{(Y-h)^2}{t}\Big)\exp\Big(-\frac{(Y-h)^2}{2t}\Big)\Big]; \tag{6}$$

the differentiations under the expectation are justified because the integrands and their derivatives in h are bounded.

Lemma 6. *Let $t \geq 600v$. Then $g(h) < g(0)$ for every h with $|h| > \sqrt{t}/2$.*

Proof. By Jensen's inequality, $g(0) \geq \exp(-v/(2t)) \geq e^{-1/1200}$. Let $|h| > \sqrt{t}/2$. On the event $\{|Y-h| \geq \sqrt{t}/4\}$ the exponential in the definition of $g(h)$ is at most $e^{-1/32}$. On the complementary event, $|Y| > |h| - \sqrt{t}/4 > \sqrt{t}/4$, and by Chebyshev's inequality this has probability at most $16v/t \leq 16/600$. Hence

$$g(h) \leq e^{-1/32} + \frac{16}{600} < 0.9960 < e^{-1/1200} \leq g(0). \qquad \square$$

Lemma 7. *Let $t \geq 600v$. Then $-H'(h) > 1/2$ for every h with $|h| \leq \sqrt{t}/2$.*

Proof. For $z \geq 0$ we have $(1-z)e^{-z/2} \geq 1 - 3z/2$: if $z \leq 1$, this follows from $e^{-z/2} \geq 1 - z/2$ and $(1-z)(1-z/2) \geq 1 - 3z/2$; if $z > 1$, then $(1-z)e^{-z/2} \geq 1 - z \geq 1 - 3z/2$. Applying this to $z = (Y-h)^2/t$ in (6) and using $\mathbb{E}(Y-h)^2 = v + h^2$ gives

$$-H'(h) \geq 1 - \frac{3(v+h^2)}{2t} \geq 1 - \frac{3}{8} - \frac{3v}{2t} \geq \frac{5}{8} - \frac{1}{400} > \frac{1}{2}. \qquad \square$$

Proof of Theorem 1. Since g is continuous and $g(h) \to 0$ as $|h| \to \infty$ by dominated convergence, g attains its maximum. By Lemma 6 every global maximizer lies in $[-\sqrt{t}/2, \sqrt{t}/2]$, and it is a zero of H. By Lemma 7, H is strictly decreasing on this interval, so it has at most one zero there. Hence the global maximizer is unique; denote it by h_t. By Lemma 7 and the mean value theorem,

$$|h_t| \leq 2\,|H(0) - H(h_t)| = 2\,|H(0)|. \tag{7}$$

Since $\mathbb{E}Y = 0$ and $0 \leq 1 - e^{-z} \leq \min\{1, z\}$ for $z \geq 0$,

$$|H(0)| = \Big|\mathbb{E}\Big[Y\Big(1 - e^{-Y^2/(2t)}\Big)\Big]\Big| \leq \mathbb{E}\Big[|Y|\min\Big\{1, \frac{Y^2}{2t}\Big\}\Big]. \tag{8}$$

The minimum is at most $|Y|/\sqrt{2t}$, which gives $|H(0)| \leq v/\sqrt{2t}$ and $|h_t| \leq \sqrt{2}\,v/\sqrt{t}$. It is also at most $Y^2/(2t)$, which gives $|H(0)| \leq \beta_3/(2t)$ and $|h_t| \leq \beta_3/t$. $\square$

3. The free case

We recall the description of $\mu \boxplus S(0,t)$ due to Biane [2], in the form given in [8, Section 2.2]. For $u \in \mathbb{R}$ and $R > 0$ let

$$F(u,R) = \mathbb{E}\Big[\frac{1}{R + (u-X)^2}\Big],$$

and let

$$v_t(u) = \inf\Big\{w \geq 0 :\ \mathbb{E}\Big[\frac{1}{w^2 + (u-X)^2}\Big] \leq \frac{1}{t}\Big\}.$$

The function v_t is continuous on $\mathbb{R}$, and whenever $v_t(u) > 0$ the number $R = v_t(u)^2$ is the unique positive solution of $F(u, R) = 1/t$ [8, (2.8)–(2.10)]. The map

$$\psi_t(u) = u + t\,\mathbb{E}\Big[\frac{u - X}{(u - X)^2 + v_t(u)^2}\Big]$$

is a homeomorphism of $\mathbb{R}$ onto $\mathbb{R}$, the measure $\mu \boxplus S(0, t)$ has a continuous density p_t, and

$$p_t(\psi_t(u)) = \frac{v_t(u)}{\pi t}, \qquad u \in \mathbb{R} \tag{9}$$

[8, (2.13) and (2.14)]. Consequently the global maximizers of p_t are the images under ψ_t of the global maximizers of v_t.

For each u the function $R \mapsto F(u, R)$ is continuous and strictly decreasing on $(0, \infty)$, and $F(u, R) \le 1/R$. As in Section 2 we assume $m = 0$, so that $Y = X$.

Lemma 8. *Let $t > v$. The function v_t attains its maximum, and $R^* := \max_u v_t(u)^2$ satisfies $t - v \le R^* \le t$. The set of global maximizers of v_t is $\{u \in \mathbb{R} : F(u, R^*) = 1/t\}$, and $F(u, R^*) \le 1/t$ for every $u \in \mathbb{R}$.*

Proof. The function $x \mapsto (R+x)^{-1}$ is convex on $[0, \infty)$, so Jensen's inequality gives $F(0, t-v) \ge (t - v + \mathbb{E}Y^2)^{-1} = 1/t$. Hence $v_t(0) > 0$, and since $F(0, \cdot)$ is decreasing, $v_t(0)^2 \ge t - v$. For fixed $\delta > 0$, dominated convergence gives $F(u, \delta^2) \to 0$ as $|u| \to \infty$, so $v_t(u) \le \delta$ for all large $|u|$. Thus the continuous function v_t tends to 0 at infinity and is positive at 0, and it attains its maximum, with $R^* \ge t - v$. At a maximizer u^* we have $1/t = F(u^*, R^*) \le 1/R^*$, so $R^* \le t$.

Let $u \in \mathbb{R}$. If $v_t(u) > 0$, then $v_t(u)^2 \le R^*$ and $F(u, R^*) \le F(u, v_t(u)^2) = 1/t$, with equality if and only if $v_t(u)^2 = R^*$. If $v_t(u) = 0$, then $F(u, w^2) \le 1/t$ for every $w > 0$, and $F(u, R^*) < 1/t$ because $F(u, \cdot)$ is strictly decreasing. This proves the last two assertions. □

Let $B = (2 + \sqrt{2})\sqrt{v}$, so that $B^2 + v = av$ with $a = 7 + 4\sqrt{2}$.

Lemma 9. *Let $t \ge 4v$. Every global maximizer u^* of v_t satisfies $|u^*| \le B$.*

Proof. Since $F(u^*, R^*) = 1/t$ and $R^* \ge t - v$,

$$\mathbb{E}\Big[\frac{(u^* - Y)^2}{R^* + (u^* - Y)^2}\Big] = 1 - R^* F(u^*, R^*) = 1 - \frac{R^*}{t} \le \frac{v}{t}. \tag{10}$$

By Chebyshev's inequality, $\mathbb{P}(|Y| \le \sqrt{2v}) \ge 1/2$. Suppose that $|u^*| > \sqrt{2v}$ and put $b = |u^*| - \sqrt{2v}$. On the event $\{|Y| \le \sqrt{2v}\}$ we have $|u^* - Y| \ge b$, and since $x \mapsto x/(R^* + x)$ is increasing and $R^* \le t$, the integrand in (10) is at least $b^2/(t + b^2)$ there. Hence $b^2/(2(t + b^2)) \le v/t$, that is, $b^2(t - 2v) \le 2vt$. For $t \ge 4v$ this gives $b^2 \le 4v$ and $|u^*| \le (2 + \sqrt{2})\sqrt{v}$. □

Lemma 10. *Let $t > 7av$. The function $u \mapsto F(u, R^*)$ is strictly concave on $[-B, B]$.*

Proof. For fixed $R > 0$ the function $u \mapsto (R + (u - x)^2)^{-1}$ and its first two derivatives are bounded uniformly in x, so $F(\cdot, R)$ is twice continuously differentiable and

$$\partial_u^2 F(u, R) = -2\,\mathbb{E}\Big[\frac{R - 3(u - Y)^2}{(R + (u - Y)^2)^3}\Big].$$

For $z \ge 0$,

$$\frac{1 - 3z}{(1 + z)^3} - (1 - 6z) = \frac{z^2(15 + 17z + 6z^2)}{(1 + z)^3} \ge 0.$$

With $z = (u - Y)^2/R$ and $\mathbb{E}(u - Y)^2 = v + u^2 \le av$ for $|u| \le B$ this gives

$$\mathbb{E}\Big[\frac{R - 3(u - Y)^2}{(R + (u - Y)^2)^3}\Big] \ge \frac{1}{R^2}\Big(1 - \frac{6av}{R}\Big).$$

If $t > 7av$, then $R^* \geq t - v > 6av$ by Lemma 8, and $\partial_u^2 F(u, R^*) < 0$ for $|u| \leq B$. □

Proof of Theorem 2. Let $t > 7av$. By Lemma 8 the global maximizers of v_t are the points at which $F(\cdot, R^*)$ attains its maximum value $1/t$; by Lemma 9 they all lie in $[-B, B]$; and by Lemma 10 the function $F(\cdot, R^*)$ has at most one maximizer in this interval. Hence v_t has a unique global maximizer u^*, and p_t has the unique global maximizer $M_{\mathrm{f}}(t) = \psi_t(u^*)$.

Write $q = u^* - Y$. Since u^* maximizes the differentiable function $F(\cdot, R^*)$, we have $\partial_u F(u^*, R^*) = 0$, that is,

$$\mathbb{E}\Big[\frac{q}{(1 + q^2/R^*)^2}\Big] = 0. \tag{11}$$

As $\mathbb{E}q = u^*$, this gives

$$u^* = \mathbb{E}\Big[q\Big(1 - \frac{1}{(1 + q^2/R^*)^2}\Big)\Big]. \tag{12}$$

Moreover, by the definition of ψ_t,

$$M_{\mathrm{f}}(t) = u^* + \frac{t}{R^*}\,\mathbb{E}\Big[\frac{q}{1 + q^2/R^*}\Big] = u^* + \frac{t}{R^*}\Big(u^* - \mathbb{E}\Big[\frac{q^3}{R^* + q^2}\Big]\Big). \tag{13}$$

Since $v < t/7$, Lemma 8 gives $R^* \geq 6t/7$, and $\mathbb{E}q^2 = v + (u^*)^2 \leq av$ by Lemma 9.

For $z \geq 0$ we have $0 \leq 1 - (1 + z)^{-2} \leq \min\{1, 2z\} \leq \sqrt{2z}$. Hence (12) gives $|u^*| \leq \sqrt{2}\,\mathbb{E}q^2/\sqrt{R^*} \leq \sqrt{2}\,av/\sqrt{R^*}$. Since $R^* + q^2 \geq 2\sqrt{R^*}|q|$, we also have $|\mathbb{E}[q^3/(R^* + q^2)]| \leq \mathbb{E}q^2/(2\sqrt{R^*}) \leq av/(2\sqrt{R^*})$. By (13),

$$|M_{\mathrm{f}}(t)| \leq \Big(1 + \frac{7}{6}\Big)\frac{\sqrt{2}\,av}{\sqrt{R^*}} + \frac{7}{6} \cdot \frac{av}{2\sqrt{R^*}} \leq \Big(\frac{13\sqrt{2}}{6} + \frac{7}{12}\Big)\sqrt{\frac{7}{6}}\,\frac{av}{\sqrt{t}} < \frac{4av}{\sqrt{t}}.$$

Let now $\beta_3 < \infty$. From $|x - y|^3 \leq 4(|x|^3 + |y|^3)$ and $|u^*| \leq B$ we get $\mathbb{E}|q|^3 \leq K := 4(B^3 + \beta_3)$. Using $1 - (1 + z)^{-2} \leq 2z$ in (12) gives

$$|u^*| \leq \frac{2K}{R^*}, \tag{14}$$

and $|\mathbb{E}[q^3/(R^* + q^2)]| \leq K/R^*$. By (13),

$$|M_{\mathrm{f}}(t)| \leq \Big(1 + \frac{7}{6}\Big)\frac{2K}{R^*} + \frac{7}{6} \cdot \frac{K}{R^*} = \frac{11K}{2R^*} \leq \frac{77K}{12\,t} < \frac{26(B^3 + \beta_3)}{t}.$$ □

Remark 11. The proof shows that the global maximizer of p_t is unique for $t > 7av$ whether or not $\mu \boxplus S(0, t)$ is unimodal. The density may have further local maxima at such times; an example is given in Section 6.

4. Expansions

We keep the assumption $m = 0$, assume $\beta_3 < \infty$, as in both parts of Theorem 3, and write $s = 1/t$. In this section $O(\cdot)$ and $o(\cdot)$ refer to $s \to 0$, with constants depending on μ.

4.1. **The classical maximizer.** Let $h = h_t$ be the maximizer of Section 2 and put $q = Y - h$. By Theorem 1, $|h| \leq \beta_3 s$, so $|q| \leq |Y| + 1$ for small s. Since $\mathbb{E}Y = 0$,

$$\mathbb{E}q = -h, \qquad \mathbb{E}q^3 = \mu_3 - 3vh - h^3, \qquad \mathbb{E}q^5 = \mu_5 - 5\mu_4 h + 10\mu_3 h^2 - 10vh^3 - h^5, \tag{15}$$

where the last identity requires $\beta_5 < \infty$. The equation $H(h) = 0$ reads $\mathbb{E}[q\,e^{-sq^2/2}] = 0$.

Proof of (a). For $x \geq 0$ write $e^{-x} = 1 - x + \rho(x)$ with $0 \leq \rho(x) \leq \min\{x, x^2/2\}$. Then

$$0 = -h - \frac{s}{2}\,\mathbb{E}q^3 + \mathbb{E}\big[q\,\rho(sq^2/2)\big], \qquad \big|\mathbb{E}\big[q\,\rho(sq^2/2)\big]\big| \leq s\,\mathbb{E}\Big[|q|^3 \min\Big\{\frac{1}{2}, \frac{sq^2}{8}\Big\}\Big].$$

The last integrand is at most $(|Y|+1)^3/2$, which is integrable, and it tends to 0 pointwise as $s \to 0$; by dominated convergence the last expectation is $o(1)$. Since $\mathbb{E}q^3 = \mu_3 + O(s)$, we get $h = -\mu_3 s/2 + o(s)$.

Proof of (b). Let $\beta_7 < \infty$. For $x \geq 0$, $e^{-x} = 1 - x + x^2/2 + \rho_3(x)$ with $|\rho_3(x)| \leq x^3/6$. Since $\mathbb{E}|q|^7$ is bounded for small s, the equation $\mathbb{E}[q\, e^{-sq^2/2}] = 0$ gives

$$0 = \mathbb{E}q - \frac{s}{2}\,\mathbb{E}q^3 + \frac{s^2}{8}\,\mathbb{E}q^5 + O(s^3).$$

Inserting (15) and $h = O(s)$,

$$h = -\frac{s}{2}\,(\mu_3 - 3vh) + \frac{s^2}{8}\,\mu_5 + O(s^3).$$

Hence $h = -\mu_3 s/2 + O(s^2)$, and substituting this into the right-hand side,

$$h = -\frac{\mu_3}{2}\,s + \frac{\mu_5 - 6v\mu_3}{8}\,s^2 + O(s^3),$$

which is (2).

4.2. **The free maximizer.** Let $t > 7av$, let $u = u^*$ and R^* be as in Section 3, and put $r = R^*/t$ and $q = u - Y$. By Lemma 8, $1 - vs \leq r \leq 1$, and by (14), $u = O(s)$. Since $|q| \leq |Y| + B$, the moments $\mathbb{E}|q|^k$ with $k \leq 7$ are bounded whenever $\beta_k < \infty$. Since $\mathbb{E}Y = 0$,

$$\mathbb{E}q = u, \quad \mathbb{E}q^2 = v + u^2, \quad \mathbb{E}q^3 = u^3 + 3vu - \mu_3, \quad \mathbb{E}q^5 = u^5 + 10vu^3 - 10\mu_3 u^2 + 5\mu_4 u - \mu_5, \tag{16}$$

where the last identity requires $\beta_5 < \infty$, and $\mathbb{E}q^4 = \mu_4 + O(s)$ if $\beta_4 < \infty$. In terms of r and s, the equation $F(u, R^*) = 1/t$, the stationarity condition (11) and the formula (13) read

$$\mathbb{E}\Big[\frac{1}{r + sq^2}\Big] = 1, \qquad \mathbb{E}\Big[\frac{q}{(r + sq^2)^2}\Big] = 0, \qquad M_{\mathrm{f}}(t) = u + \mathbb{E}\Big[\frac{q}{r + sq^2}\Big]. \tag{17}$$

We use the following expansions, valid for $z \geq 0$:

$$\frac{1}{1+z} = 1 - z + \rho_1(z) = 1 - z + z^2 + \sigma_1(z), \qquad 0 \leq \rho_1(z) \leq \min\{z, z^2\}, \qquad |\sigma_1(z)| \leq z^3, \tag{18}$$

$$\frac{1}{(1+z)^2} = 1 - 2z + \rho_2(z) = 1 - 2z + 3z^2 + \sigma_2(z), \quad 0 \leq \rho_2(z) \leq \min\{2z, 3z^2\}, \quad |\sigma_2(z)| \leq 4z^3. \tag{19}$$

Here $\rho_1(z) = z^2/(1+z)$ and $\sigma_1(z) = -z^3/(1+z)$; the bounds on ρ_2 and σ_2 follow from Taylor's formula, since the second and third derivatives of $(1+z)^{-2}$ are $6(1+z)^{-4}$ and $-24(1+z)^{-5}$, together with $(1+z)^{-2} \leq 1$. Throughout we take $z = sq^2/r$.

Proof of (a). Multiplying the second equation in (17) by r^2 and using (19),

$$0 = u - \frac{2s}{r}\,\mathbb{E}q^3 + \mathbb{E}\big[q\,\rho_2(z)\big], \qquad \big|\mathbb{E}\big[q\,\rho_2(z)\big]\big| \leq \frac{s}{r}\,\mathbb{E}\Big[|q|^3 \min\Big\{2, \frac{3sq^2}{r}\Big\}\Big] = o(s),$$

by dominated convergence as in the classical case. Since $\mathbb{E}q^3 = -\mu_3 + O(s)$ and $1/r = 1 + O(s)$, this gives $u = -2\mu_3 s + o(s)$. In the same way, the third relation in (17) and (18) give

$$M_{\mathrm{f}}(t) = u + \frac{1}{r}\Big(u - \frac{s}{r}\,\mathbb{E}q^3 + \mathbb{E}\big[q\,\rho_1(z)\big]\Big) = 2u + \mu_3 s + o(s) = -3\mu_3 s + o(s).$$

Proof of (b). Let $\beta_7 < \infty$. Multiplying the first equation in (17) by r and using (18),

$$r = 1 - \frac{s}{r}\,\mathbb{E}q^2 + \frac{s^2}{r^2}\,\mathbb{E}q^4 + O(s^3).$$

With (16) this gives $r = 1 - vs + O(s^2)$ and then

$$r = 1 - vs + (\mu_4 - v^2)s^2 + O(s^3), \qquad \frac{1}{r} = 1 + vs + O(s^2), \qquad \frac{1}{r^2} = 1 + 2vs + O(s^2).$$

Multiplying the second equation in (17) by r^2 and using (19),

$$0 = \mathbb{E}q - \frac{2s}{r}\,\mathbb{E}q^3 + \frac{3s^2}{r^2}\,\mathbb{E}q^5 + O(s^3) = u - 6vsu + 2\mu_3 s + 2v\mu_3 s^2 - 3\mu_5 s^2 + O(s^3),$$

where we inserted (16) and $u = O(s)$. Hence $u = -2\mu_3 s + O(s^2)$ and then

$$u = -2\mu_3 s + (3\mu_5 - 14v\mu_3)s^2 + O(s^3).$$

Finally, by the third relation in (17) and (18),

$$\begin{aligned} M_{\mathrm{f}}(t) &= u + \frac{1}{r}\,\mathbb{E}q - \frac{s}{r^2}\,\mathbb{E}q^3 + \frac{s^2}{r^3}\,\mathbb{E}q^5 + O(s^3) \\ &= 2u - 2vsu + \mu_3 s + (2v\mu_3 - \mu_5)s^2 + O(s^3) \\ &= -3\mu_3 s + (5\mu_5 - 22v\mu_3)s^2 + O(s^3), \end{aligned}$$

which is (3). This completes the proof of Theorem 3.

The forms (4) and (5) follow by expanding $1/(t+v)$ and $1/(t+v)^2$ in powers of s: the coefficients of s^2 are $v\mu_3/2 + (\mu_5 - 10v\mu_3)/8 = (\mu_5 - 6v\mu_3)/8$ and $3v\mu_3 + 5(\mu_5 - 5v\mu_3) = 5\mu_5 - 22v\mu_3$. The cumulants and free cumulants used there are those of a centred law: $\kappa_5 = \mu_5 - 10\mu_2\mu_3$, and $r_5 = \mu_5 - 5\mu_2\mu_3$ [15, Lecture 11]; the normal law has no cumulants of order higher than two, and the semicircle law has no free cumulants of order higher than two.

5. Proofs of the corollaries

Proof of Corollary 4. By Jensen's inequality, $\exp(\varepsilon\,\mathbb{E}X^2) \le \alpha$, so $v \le \mathbb{E}X^2 \le L$ and $m^2 \le L$. For $k > 0$ the function $x \mapsto x^k e^{-\varepsilon x^2}$ on $[0,\infty)$ has the maximum $(k/(2e\varepsilon))^{k/2}$, hence

$$\mathbb{E}|X|^k \le \alpha\Big(\frac{k}{2e\varepsilon}\Big)^{k/2}.$$

In particular all moments of μ are finite, and $\beta_3 \le 4(\mathbb{E}|X|^3 + |m|^3) \le 8\,\mathbb{E}|X|^3 \le Q$. Since μ is not a point mass, $v > 0$.

(i) For $t \ge 600L$, Theorem 1 applies and gives a unique global maximizer $M_{\mathrm{c}}(t)$ of the density with $|M_{\mathrm{c}}(t) - m| \le Q/t$. For $t \ge 36\log(2\alpha)/\varepsilon$ the law $\mu * N(0,t)$ is unimodal by [8, Theorem 4.3]. The density of a unimodal law is maximal at each of its modes, so the mode is unique and equal to $M_{\mathrm{c}}(t)$.

Now let $t \ge 72\log(2\alpha^2)/\varepsilon$, and let $Y = X - m$. Since $Y^2 \le 2X^2 + 2m^2$ and $e^{\varepsilon m^2} \le \alpha$, we have $\mathbb{E}e^{\varepsilon Y^2/2} \le \alpha^2$, so the law of Y satisfies (1) with $\varepsilon/2$ in place of ε and a constant at most α^2 in place of α. By [8, Theorem 4.3] and the statement on the mode in [8, Problem 4.6], applied to the law of Y, the law $\mu * N(0,t)$ is unimodal and each of its modes has the form $m + h$ with $|h| \le \sqrt{t}/2$. The density is maximal at each mode, so $H(h) = 0$. Since $t \ge 144L \ge 36v$, the first inequality in the proof of Lemma 7 gives

$$-H'(h) \ge 1 - \frac{3}{8} - \frac{3v}{2t} \ge \frac{7}{12} \qquad (|h| \le \sqrt{t}/2),$$

so H has only one zero in $[-\sqrt{t}/2, \sqrt{t}/2]$ and the mode is unique. By the mean value theorem and (8), $|h| \le \frac{12}{7}|H(0)| \le \frac{6}{7}\beta_3/t \le Q/t$.

(ii) and (iii) follow from Theorems 2 and 3, since $7av \le 7aL$ and $v^{3/2} \le L^{3/2}$. □

Proof of Corollary 5. Let $[c-D/2, c+D/2]$ be the smallest interval containing the support of μ. Then $v \le \mathbb{E}(X-c)^2 \le D^2/4$, and $|Y| \le D$ gives $\beta_3 \le Dv \le D^3/4$. Hence $7av \le 7aD^2/4 < 23D^2$, and for $t \ge 23D^2$ Theorem 2 gives a unique global maximizer $M_{\mathrm{f}}(t)$ of p_t with

$$|M_{\mathrm{f}}(t) - m| \le \frac{26}{t}\Big(\frac{(2+\sqrt{2})^3}{8} + \frac{1}{4}\Big)D^3 < \frac{136\,D^3}{t}.$$

For $t \ge 4D^2$ the law $\mu \boxplus S(0,t)$ is unimodal by [8, Theorem 3.2], and as in the proof of Corollary 4 its mode is unique and equal to $M_{\mathrm{f}}(t)$. All moments of μ are finite, so Theorem 3 applies. □

6. An example and numerical checks

Let $\mu = \frac{2}{3}\delta_{-1} + \frac{1}{3}\delta_2$. Then $m = 0$, $v = 2$, $\mu_3 = 2$ and $\mu_5 = 10$, and Theorem 3 gives

$$M_{\mathrm{c}}(t) = -\frac{1}{t} - \frac{7}{4t^2} + O(t^{-3}), \qquad M_{\mathrm{f}}(t) = -\frac{6}{t} - \frac{38}{t^2} + O(t^{-3}).$$

Table 1 compares these with the maximizers computed from the equations $H(h) = 0$ and (17).

t	$t\,M_{\mathrm{c}}(t)$	$-1 - 7/(4t)$	$t\,M_{\mathrm{f}}(t)$	$-6 - 38/t$
10^4	-1.00017502375	-1.000175	-6.00380137932	-6.0038
10^5	-1.00001750024	-1.0000175	-6.00038001380	-6.00038
10^6	-1.00000175000	-1.00000175	-6.00003800014	-6.000038

Table 1. The global maximizers for $\mu = \frac{2}{3}\delta_{-1} + \frac{1}{3}\delta_2$ and their two-term expansions.

The program `code/mode_checks.py` in the repository of the paper derives the coefficients in (2)–(5) symbolically from the equations $H(h) = 0$ and (17), using SymPy [12]. With mpmath [14] it solves the equations $H(h) = 0$ and (17) near the mean for five laws with finitely many atoms, for t between 10^3 and 10^6, and for a centred exponential law, for t between 10^3 and 10^5; in each case the difference of the solution from the two-term expansion, multiplied by t^3, approaches a constant. For the five atomic laws it also evaluates the two densities on a grid at $t = 600v$ and at a time just above $7av$, and confirms that the global maximizer is unique and satisfies the bounds of Theorems 1 and 2, together with the inequalities of Lemmas 7, 8, 9 and 10. For the law $0.8\,\delta_0 + 0.199\,\delta_1 + 0.001\,\delta_{50}$, at a time just above $7av$, the free density has a second local maximum besides its global maximizer, so $\mu \boxplus S(0,t)$ is not unimodal at that time. A second program, `code/checks.py`, locates both global maximizers of four laws with finitely many atoms by a grid search followed by Newton's method at forty-digit precision and compares them with the expansions. The polynomial identity in the proof of Lemma 10 is also checked in Lean 4 [13] with Mathlib [11] (`lean/Algebra.lean`).

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [16] and working together inside the Brain framework [18], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [17], the Math Brain repository [16] and the Brain framework repository [18].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant

agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [18] and Math Brain [16]. In October 2026 a Codex agent (OpenAI) connected to the Brain, working from the project's list of open problems in probability, obtained the expansions of Theorem 3(b), the uniqueness of the free maximizer and bounds of order t^{-1} for both maximizers, together with numerical checks and the Lean file; further AI agents, among them Claude Opus 5.5 (Anthropic), checked that work and wrote the program `code/checks.py`. In November 2026 a Claude Fable agent (Anthropic) obtained Theorems 1, 2 and 3 and the two corollaries in their present form with their proofs, wrote and ran the programs and prepared the manuscript, and a further Codex agent, working from the manuscript alone with its own programs, checked the proofs and recomputed the coefficients and the example. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/brownian-mode-location`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [16].

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